Dimension counts independent directions

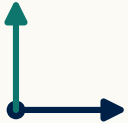
dimension 0

point

dimension 1

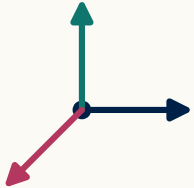
line

dimension 2



plane

dimension 3



space (sketch)